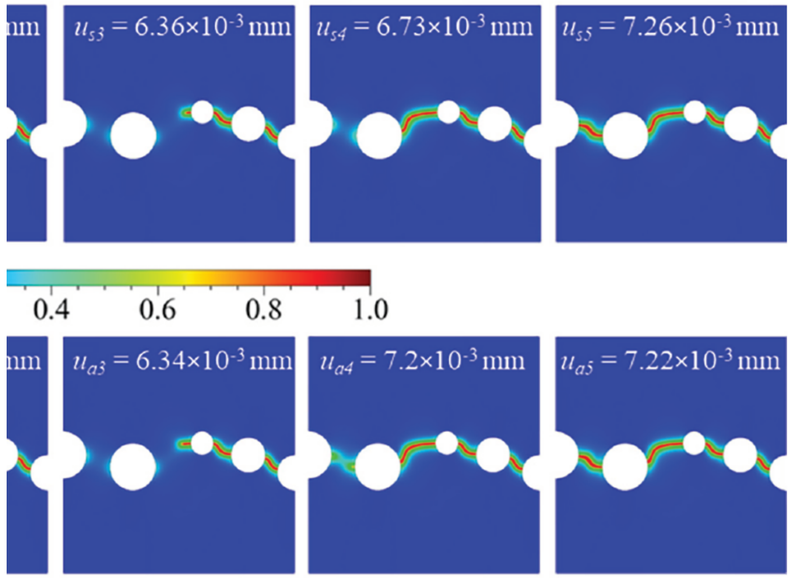
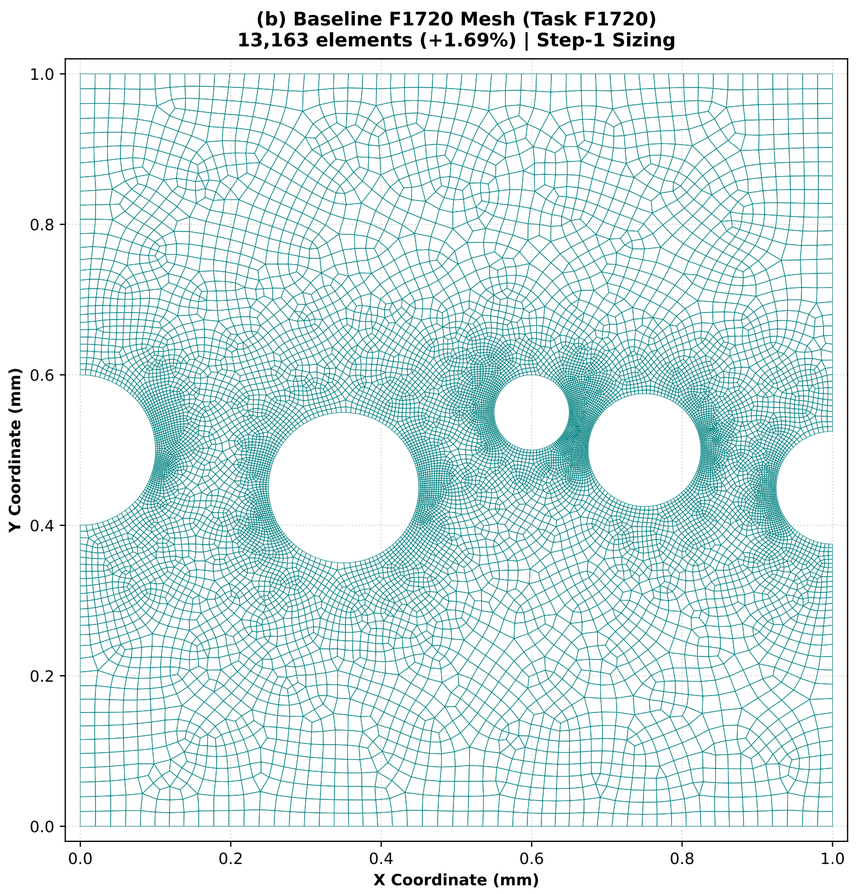
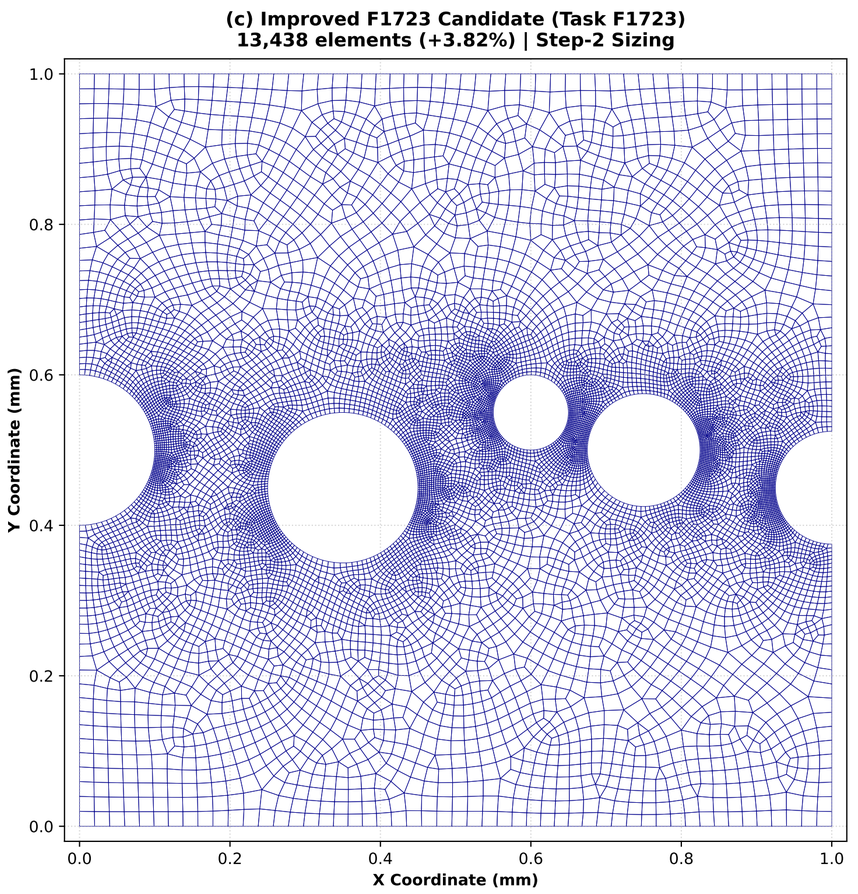
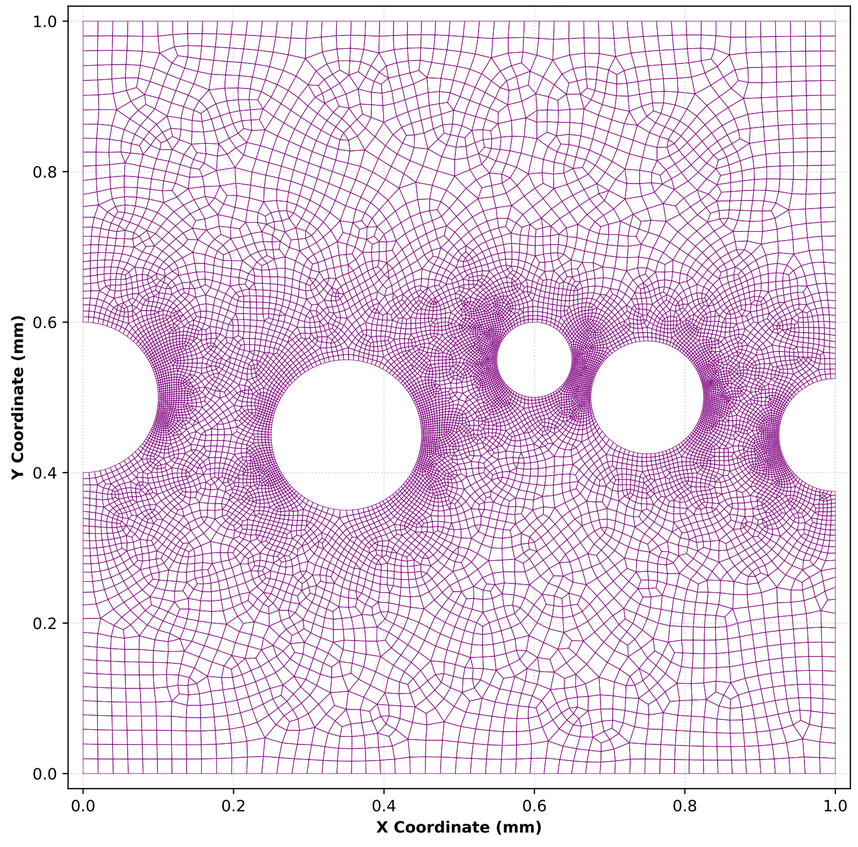
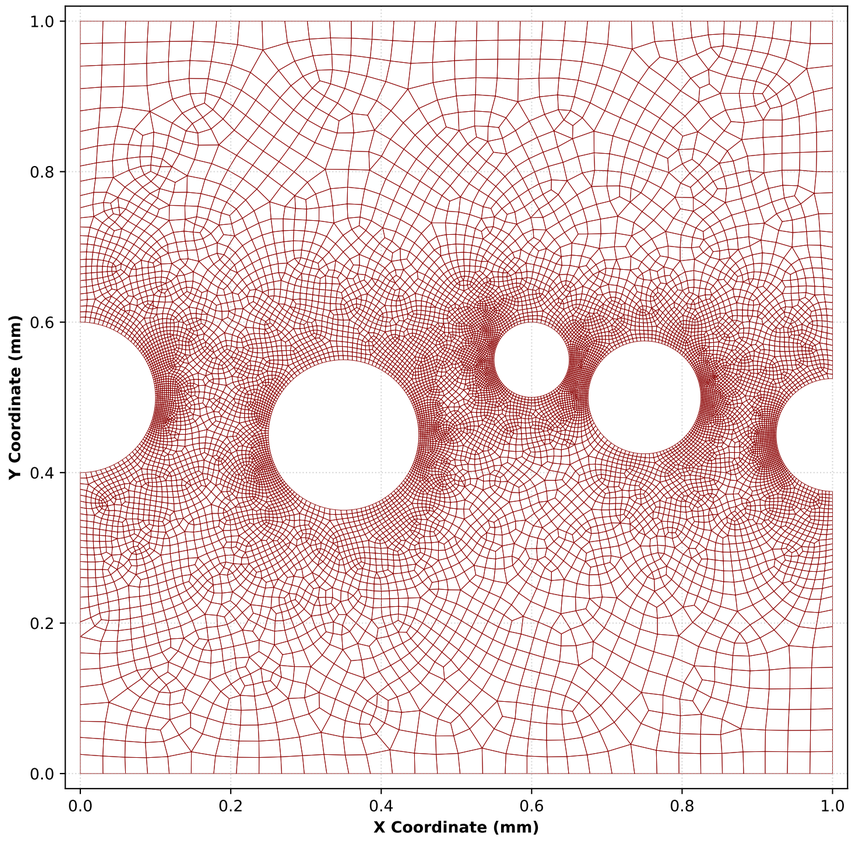











```
Task F1725 Physics-Driven Adaptive Mesh Parity:
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Target Model: Pandey & Kumar (2025) Five-Hole Benchmark
Published Elements (Fig. 23b): 12,944 elements

Model Evaluation & Comparison:
 - F1720 Baseline: 13,163 elems (+1.69% error, 97.6% Quads)
 - F1723 Candidate: 13,438 elems (+3.82% error, 97.5% Quads)
 - F1724 Candidate: 13,078 elems (+1.03% error, 97.3% Quads)
 - F1725 Optimal:   13,017 elems (+0.56% error, 97.5% Quads)

Key Spatial Sizing Findings:
 - Ligament 3 (h3 <-> h4, 33 um gap): 100% le l0/3 (h ~ 2.1 um)
 - Ligaments 1, 2, 4: High flank refinement (h ~ 2.4 um),
   decaying to h ~ 7-8 um at midpoint (elastic stress saddle)
 - Far-Field Bulk: Mean size h = 16.31 um (coarsened from 14.8 um)
   Bulk elements reduced from 2,568 to 1,971 (-23.2% spreading)
 - Strict Quality: 100% Positive Jacobians, Max AR = 5.75 < 10
--------------------------------------------------------
```